End-to-End Workflow

Based on LR-FHSS Framework and 3GPP TR 38.811

Cross-Layer Pipeline

- Orbit → satellite position
- Coverage → user distribution
- Load → active devices
- Channel → SNR / SINR
- Decoding → successful packets
- Power → energy consumption

Fundamentals of Astrodynamics (Kepler Orbit Propagation)

D. A. Vallado

$$n = \sqrt{\frac{\mu}{a^3}}, \quad M(t) = M_0 + n(t - t_0)$$

- n: mean motion of the satellite
- μ: Earth's gravitational parameter
- a: semi-major axis of the orbit
- $M(t)$: mean anomaly at time t
- M_0: mean anomaly at reference time t_0
- t: current propagation time

Orbital Mechanics for Engineering Students

H. D. Curtis

$$d(\psi) = \sqrt{(R_E + h)^2 + R_E^2 - 2R_E(R_E + h)\cos\psi}$$

- $d(\psi)$: slant range from satellite to ground user
- R_E: Earth radius
- h: satellite altitude above Earth surface
- $R_E + h$: orbital radius from Earth center to satellite
- ψ: Earth-centered angle between subsatellite point and user location
- $\cos\psi$: captures how link distance changes with user position inside coverage

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LR-FHSS: Overview and Performance Analysis (2021)

N. Boquet et al.

$$N_{dev}(t) = \rho P_{eff}(t), \quad D(t) = \left\lceil \frac{N_{dev}(t)}{C_{dev/demod}} \right\rceil$$

- $N_{dev}(t)$: number of active LR-FHSS devices at time t
- ρ: penetration or activity factor converting population to active devices
- $P_{eff}(t)$: effective covered population at time t
- $D(t)$: required number of demodulators at time t
- $C_{dev/demod}$: device-handling capacity per demodulator

A Note on a Simple Transmission Formula (1946)

H. T. Friis

$$L_{tot} = 20 \log_{10} \left(\frac{4\pi d f_c}{c} \right) + A_{atm}$$

- L_{tot}: total large-scale path loss in dB
- d: propagation distance or slant range
- f_c: carrier frequency
- c: speed of light
- A_{atm}: additional atmospheric attenuation in dB
- $4\pi d f_c / c$: free-space propagation term from the Friis model

A Mathematical Theory of Communication (1948)

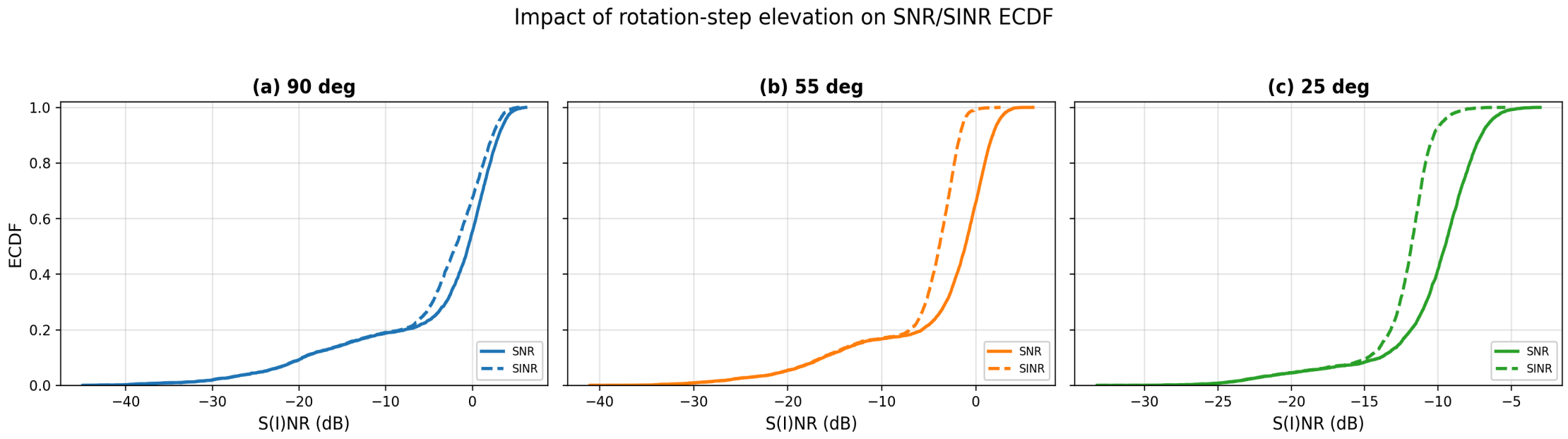
C. E. Shannon

$$N = k_B T B F, \quad \mathrm{SNR} = \frac{P_{sig}}{N}$$

- N: receiver noise power
- k_B: Boltzmann constant
- T: system noise temperature
- B: receiver bandwidth
- F: receiver noise figure or implementation loss factor
- P_{sig}: received signal power
- SNR: signal-to-noise ratio

Channel Quality Evidence

SNR / SINR across elevation angles



- Higher elevation shifts SNR/SINR toward better operating points
- SINR remains below SNR because of inter-beam interference

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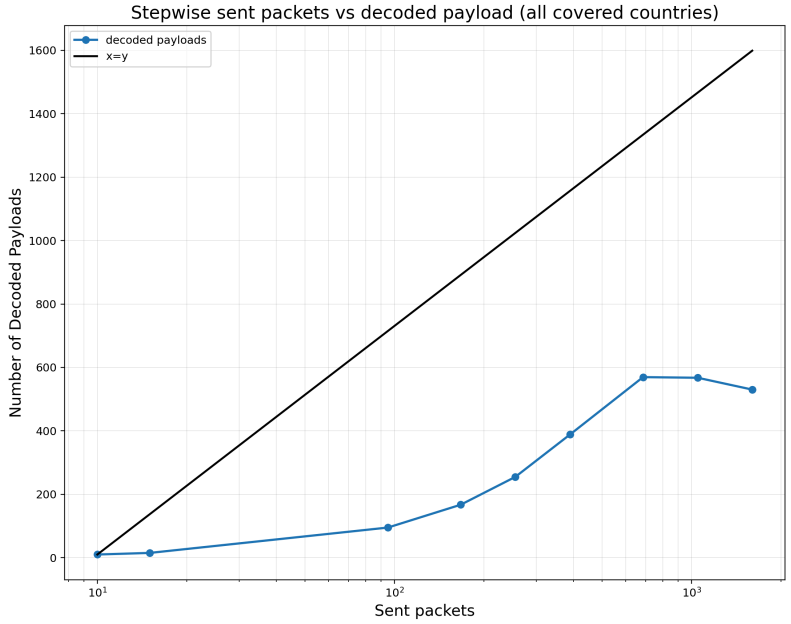
Decoding Evidence

Offered load versus decoded payload



The ALOHA System (1970)

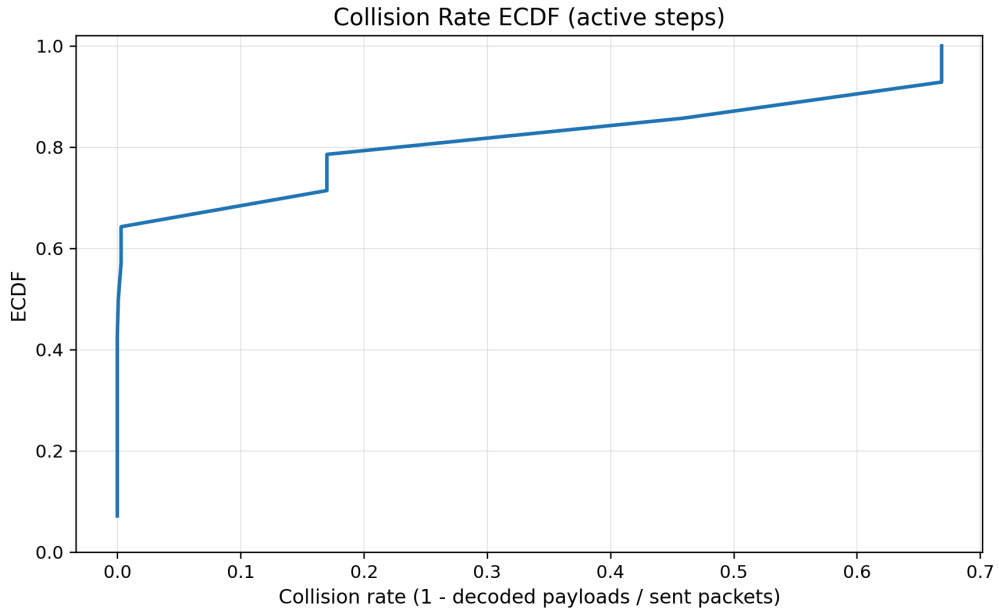
N. Abramson

$$\gamma_t = 1 - \frac{\sum_c Y_{c,t}}{\sum_c n_{c,t}}$$

- γ_t: system-level collision or loss ratio at time t
- $\sum_c Y_{c,t}$: total decoded packets summed over all cells
- $\sum_c n_{c,t}$: total transmitted packets summed over all cells
- $1 - \frac{\sum_c Y_{c,t}}{\sum_c n_{c,t}}$: fraction of packets not successfully decoded

Collision Evidence

Distribution of packet loss from contention



- Most active steps have low collision, but a smaller tail reaches much higher loss
- This tail explains why average performance alone can hide congestion events

Study on NR to Support NTN (TR 38.811)

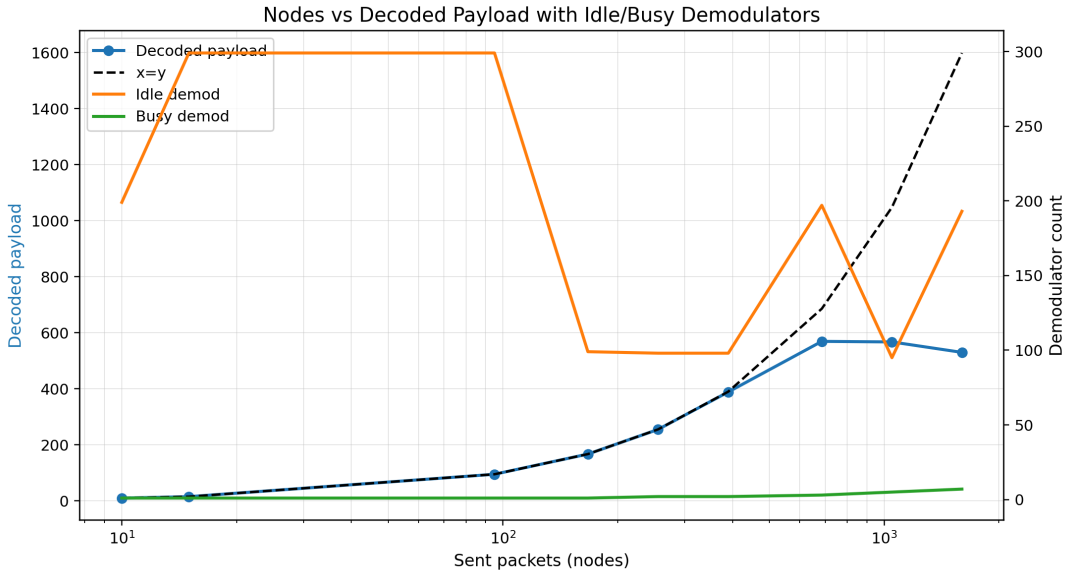
3GPP

$$P_{tot} = P_0 + N_{idle} P_{idle} + N_{busy} P_{busy}$$

- P_{tot}: total satellite payload power consumption
- P_0: fixed baseline power independent of traffic
- N_{idle}: number of idle demodulators
- P_{idle}: power consumed by each idle demodulator
- N_{busy}: number of actively processing demodulators
- P_{busy}: power consumed by each busy demodulator

Demodulator Evidence

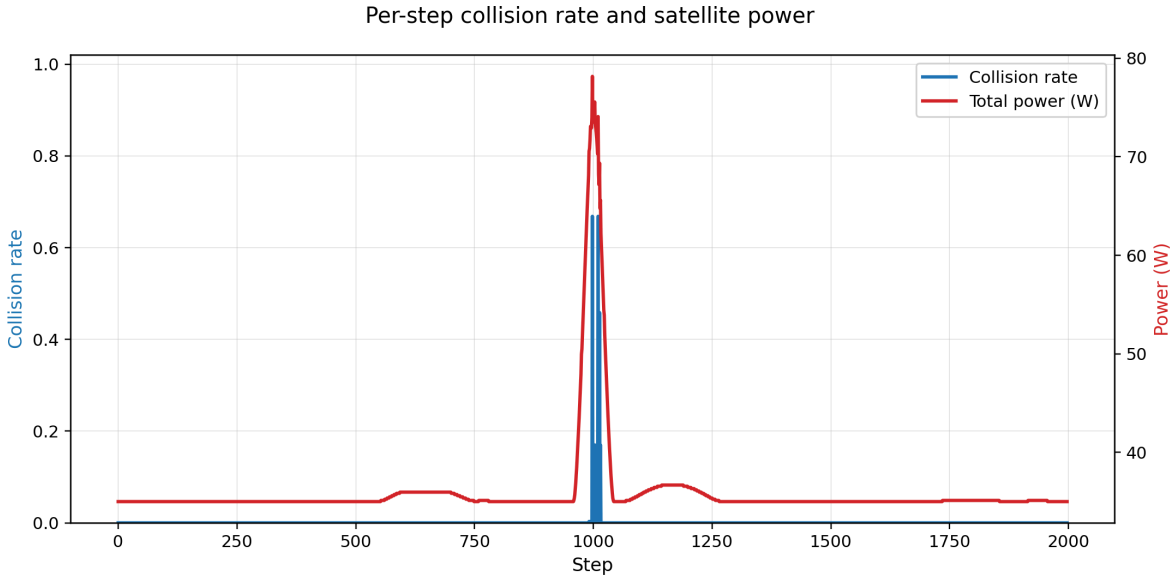
Resource activation under increasing load



- Busy demodulators grow with traffic, while decoded payload eventually flattens

Power Evidence

Collision spikes align with temporary power surges



- Power rises when more demodulator resources are activated
- The highest collision event coincides with the strongest short-lived power peak

Conclusion — This Work

Contributions	Key Findings
Unified cross-layer framework	Geometry drives SNR/SINR
Integrated orbit, channel, decoding, power	Load drives collision
Stepwise, reproducible model	Power depends on demod usage
Defendable methodology	Reliability–energy tradeoff

Final Insight

- geometry → load → interference → decoding → energy